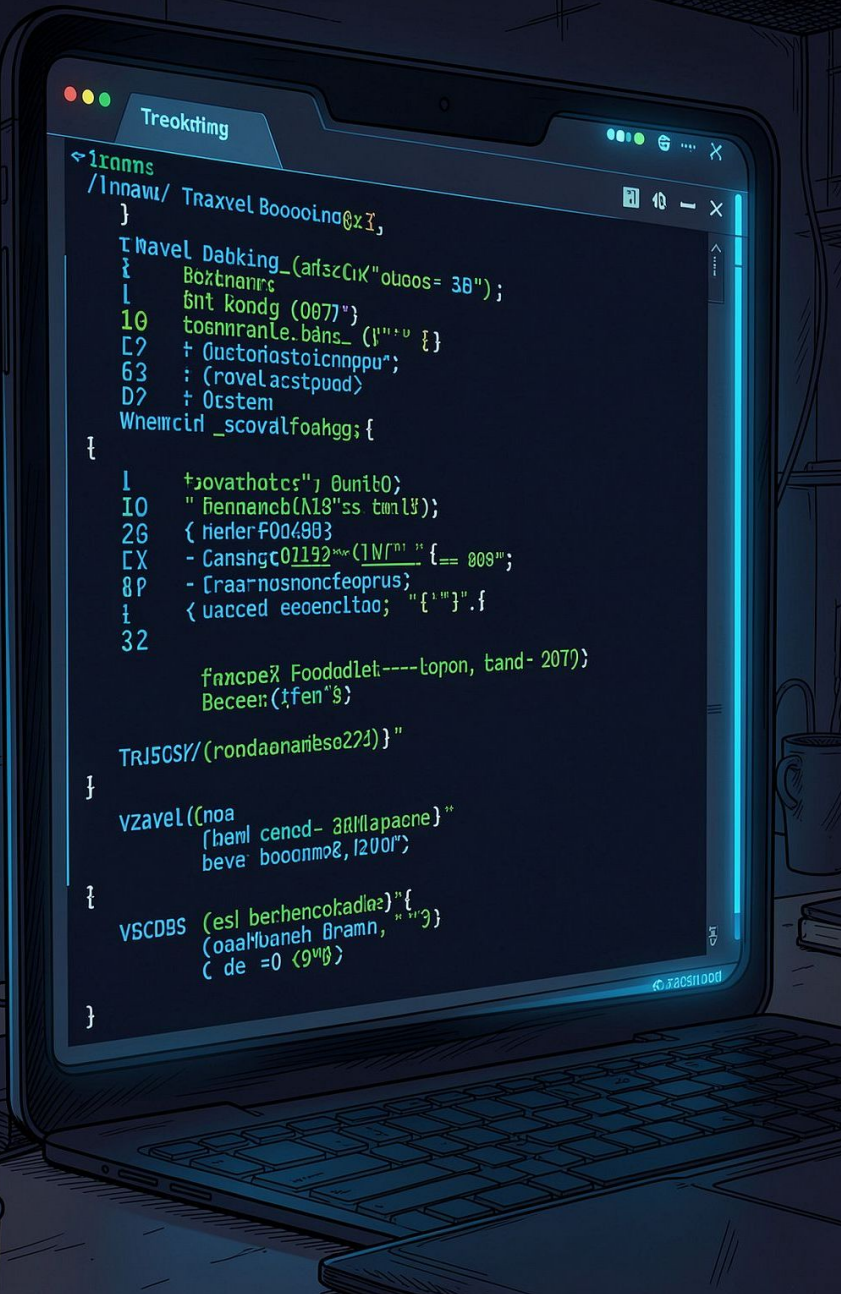




Fast-Ticket-Booker Python Project



Fast Ticket Booker

A lightweight, terminal-based CLI reservation system for inter-city travel — built in Python, storing bookings in plain text and JSON.

CLI APP

PYTHON

JSON + TXT STORAGE

What Is Fast Ticket Booker?



Fast Ticket Booker is a console-based reservation application that simplifies multi-modal inter-city travel booking. Passengers select their vehicle, route, and seating class – the system calculates dynamic pricing and archives every confirmed booking.

Fast

Minimal steps, instant confirmation

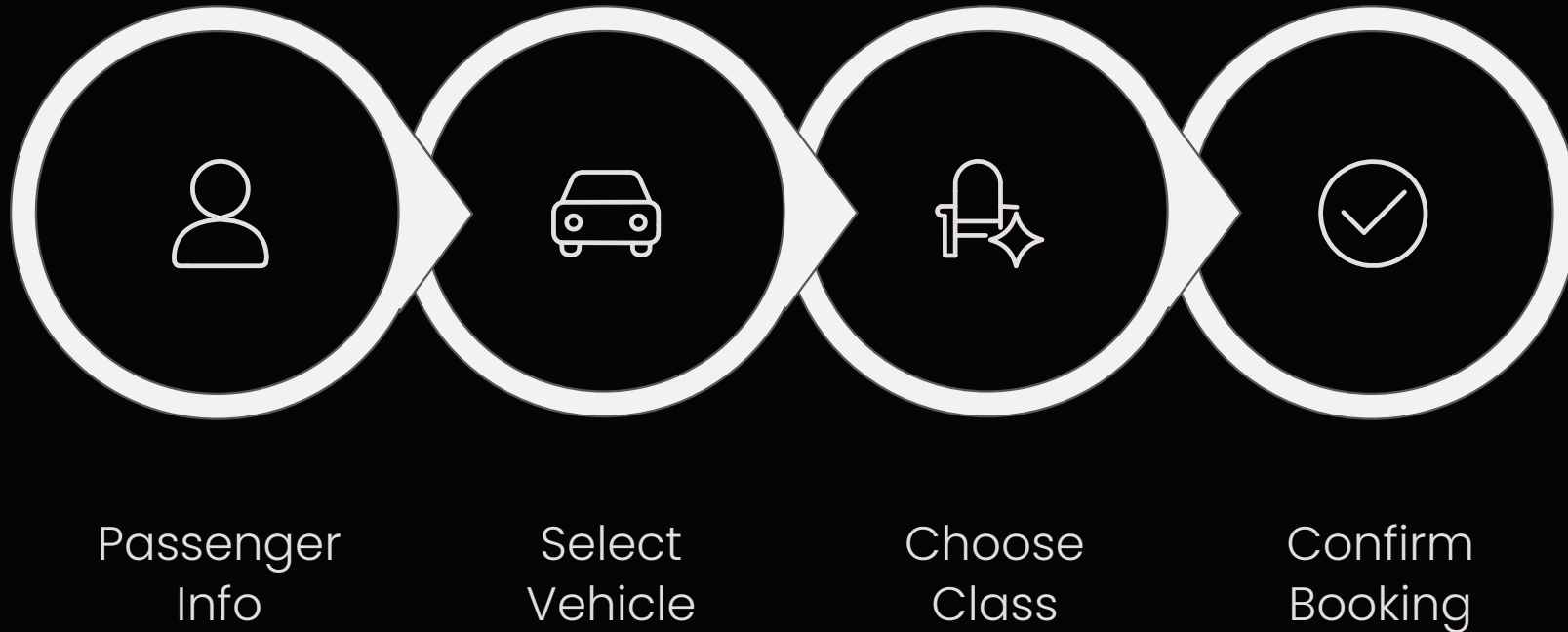
Reliable

Validated inputs, structured output

Portable

Runs in any terminal, no GUI needed

Booking Flow at a Glance



Each phase includes input validation loops. Invalid entries prompt the user to retry until valid data is provided.

Vehicle & Route Selection

Supported Fleet



Bus

Inter-city road transport



Train

Rail network — eligible for 5% discount



Car

Private or shared ride



Van

Group travel option

Route Input Rules

- **Source (From):** e.g., Lahore — empty input not allowed
- **Destination (To):** e.g., Karachi — must differ from source
- Both fields are validated before proceeding



Train bookings automatically trigger a 5% discount on all seating classes.

Seating Classes & Pricing

Every vehicle supports three standard classes. Base prices are defined in `data/prices.py` as a dictionary keyed by vehicle and class.

Class	Bus	Car / Van	Train
Economy	PKR 2,000	PKR 2,500	PKR 3,000
Business	PKR 3,500	PKR 4,000	PKR 4,500
Sleeper	PKR 5,000	PKR 6,000	PKR 6,500

 **Train Discount:** 5% off total — applied automatically after `base_price × passengers`.

Confirmation & Ticket ID



Step 6 — Verify Before Committing

Before saving, the app displays a formatted receipt preview. The user must explicitly confirm with **Yes** or **No**.

Yes

Generates a unique Ticket ID (e.g., FTB-49201), saves to both files, prints final receipt.

No

Cancels booking or returns to selection step — no data is written.

Data Archiving

Every confirmed booking is persisted to two parallel formats for human readability and programmatic access.

bookings.txt

Receipt-style plain text record. Appended line-by-line for easy terminal viewing and printing.

bookings.json

Structured JSON objects. Enables booking history retrieval, filtering, and future API integration.

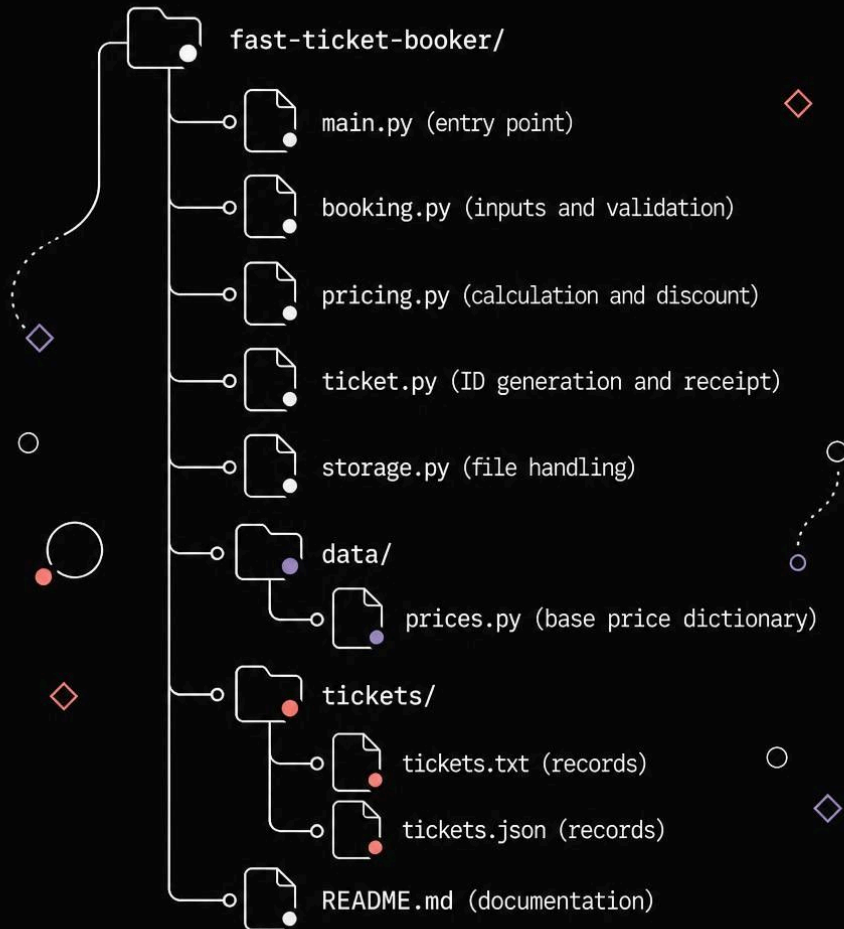
```
{  
  "ticket_id": "FTB-49201",  
  "passenger": "Zeeshan Ahmed",  
  "date": "2026-10-20",  
  "vehicle": "Train",  
  "route": "Islamabad -> Lahore",  
  "class": "Business",  
  "passengers": 2,  
  "total": 5700.00  
}
```

Sample Terminal Receipt

Output printed to console and saved to `tickets.txt` on confirmation.

```
=====
FAST TICKET BOOKER — RECEIPT
=====
Ticket ID : FTB-49201
Passenger Name : Zeeshan Ahmed
Travel Date : 2026-10-20 at 09:00 AM
Vehicle : Train
Route : Islamabad -> Lahore
Class : Business Class
Passengers : 2
Base Price : PKR 3,000 × 2 = PKR 6,000
Discount : 5% (Train Special)
Total Amount : PKR 5,700.00
=====
```

Project Structure



Module Responsibilities

- `main.py` — Orchestrates the full booking flow
- `booking.py` — Collects and validates all user inputs
- `pricing.py` — Calculates totals, applies train discount
- `ticket.py` — Generates unique IDs, formats receipts
- `storage.py` — Writes to `.txt` and `.json`, reads history

Build It Right

Fast Ticket Booker is a focused, well-structured CLI project — ideal for demonstrating input validation, dynamic pricing, file I/O, and modular Python design.

01

Start with `main.py`

Scaffold the welcome banner and main loop.

02

Implement validation in `booking.py`

Loop on invalid input; reject empty fields and non-positive integers.

03

Add pricing and storage

Wire up `pricing.py` and `storage.py`, then test end-to-end.

